Topological Band Theory: Static, Driven, and Anomalous

Where the Chern number classifies a phase, and where it stops

Amaira Goel

Department of Physics, Indian Institute of Technology Kanpur

amairag24@iitk.ac.in | github.com/amairagoel

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Abstract

This report documents three numerical studies in topological band theory, built from scratch and each validated against an analytically known result before any figure was produced. The first reproduces the Haldane model and its closed-form phase boundary. The second shows that circularly driven graphene generates the Haldane model, so the static model is the high-frequency limit of a driven one; the induced gap coefficient is measured numerically rather than quoted, since it is convention dependent. The third exhibits the anomalous Floquet phase of Rudner et al., in which both bulk Chern numbers vanish while chiral edge modes traverse the entire quasienergy zone — the static bulk–boundary correspondence fails because the Chern number is built from the one-period operator alone and discards the micromotion. None of the physics here is new; the intent is a correct, checkable implementation and an explicit account of where the standard invariant stops applying. Three genuine errors caught by the validation suites are documented rather than silently corrected.

1 The question

A gapped two-dimensional band structure carries an integer, the Chern number, computed purely from the bulk. That integer counts the chiral channels running along the sample's edge. One never looks at the edge to obtain it. This is the bulk–boundary correspondence, and the robustness of the edge channels follows from the fact that an integer cannot change continuously.

The question organising this report is narrower than "how does topology work": it is when does the invariant actually classify the phase, and what happens at the boundary of its validity? Section 3 establishes the standard picture. Section 4 shows that periodic driving reproduces it. Section 5 exhibits a driven system where it fails outright.

2 Method: the Chern number without a gauge

If the Chern number is nonzero then there is no smooth global choice of phase for the Bloch states across the Brillouin zone. That obstruction is precisely what the invariant measures, so any method requiring a smooth gauge is self-defeating.

All Chern numbers here use the Fukui–Hatsugai–Suzuki plaquette construction [2]. Link variables are built from overlaps of neighbouring states,

$$U_\mu(\mathbf{k}) = \frac{\langle u(\mathbf{k})|u(\mathbf{k}+\hat{\mu})\rangle}{|\langle u(\mathbf{k})|u(\mathbf{k}+\hat{\mu})\rangle|},$$

and the field strength is the phase of the product around each plaquette. Every arbitrary phase cancels between adjacent links, so the result is gauge invariant by construction and the sum over plaquettes is an exact integer even on a coarse grid. In practice the Haldane Chern number is quantised to 10^{-12} on a 40×40 grid.

3 The Haldane model

Haldane showed that a quantised Hall conductance does not require a magnetic field [1]. On a honeycomb lattice, complex next-nearest-neighbour hopping breaks time-reversal symmetry locally while the net flux through a unit cell vanishes. With $H(\mathbf{k}) = d_0 \mathbb{1} + \mathbf{d}(\mathbf{k}) \cdot \boldsymbol{\sigma}$ and

$$d_x + i d_y = t_1 \sum_j e^{i\mathbf{k}\cdot\boldsymbol{\delta}_j}, \qquad d_z = M - 2t_2 \sin\phi \sum_j \sin(\mathbf{k}\cdot\mathbf{a}_j),$$

the gap closes and the topology changes at $|M| = 3\sqrt{3}\, t_2 |\sin\phi|$. That closed form is the validation target.

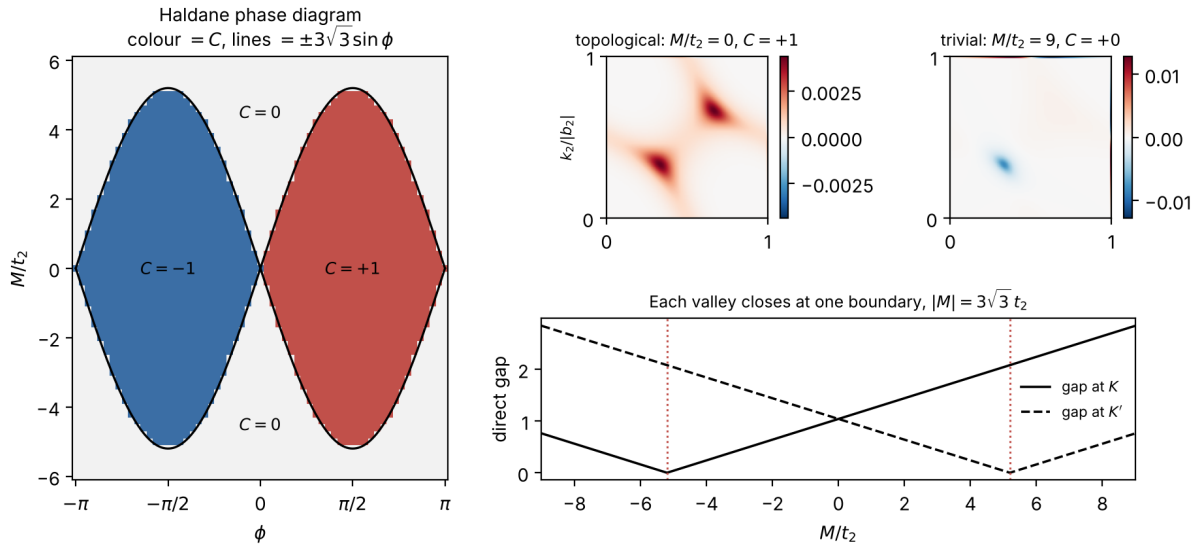


Figure 1: Haldane phase diagram. Left: Chern number over $(\phi, M/t_2)$, with the analytic boundaries $\pm 3\sqrt{3}\sin\phi$ overlaid as solid lines — computed and predicted boundaries coincide. Top right: Berry curvature concentrates at the Dirac points in the topological phase; in the trivial phase it carries both signs and cancels. Bottom right: the Haldane mass is $+3\sqrt{3}\, t_2 \sin\phi$ at K and $-3\sqrt{3}\, t_2 \sin\phi$ at K', so each valley closes at exactly one of the two boundaries.

Three corrections worth recording. Each was caught by a failing test rather than by inspection.

1. The sign of C is a convention, fixed by the orientation ordering of the neighbour vectors, not by physics. The first draft asserted $C = -1$; the implementation gives $+1$. What is physical, and what the tests now assert, is $|C| = 1$ together with $C \to -C$ under $\phi \to -\phi$.
2. Brillouin-zone sampling. The Dirac points sit at fractional coordinates $(1/3, 2/3)$ and $(2/3, 1/3)$, so a uniform $n \times n$ grid samples them exactly only when $3 \mid n$. Verifying gap closure by minimising over a grid therefore fails silently for most n: at the critical mass, $n = 300$ gives a minimum gap of 10^{-16} while $n = 400$ gives 10^{-2}. The gap must be evaluated at K and K' directly.
3. A factor of two. The familiar $3\sqrt{3}\, t_2$ is the Dirac mass, i.e. the half-gap; the full band splitting is $6\sqrt{3}\, t_2$. This propagates directly into Section 4, where it initially produced a clean factor-of-2.008 discrepancy.

4 Driving generates the Haldane model

Irradiate plain graphene with circularly polarised light. The vector potential enters by Peierls substitution, $\mathbf{k} \to \mathbf{k} + \mathbf{A}(t)$ with $\mathbf{A}(t) = A(\cos \Omega t,\, h \sin \Omega t)$ and helicity $h = \pm 1$. Graphene alone is gapless, but the drive generates an effective next-nearest-neighbour hopping with a complex phase — exactly the Haldane term — and opens a gap at the Dirac points [3].

Because the drive is periodic, states are labelled by quasienergy: the eigenphases of the one-period evolution operator $U(T)$, defined only modulo Ω. Quasienergy lives on a circle rather than a line, which becomes essential in Section 5.

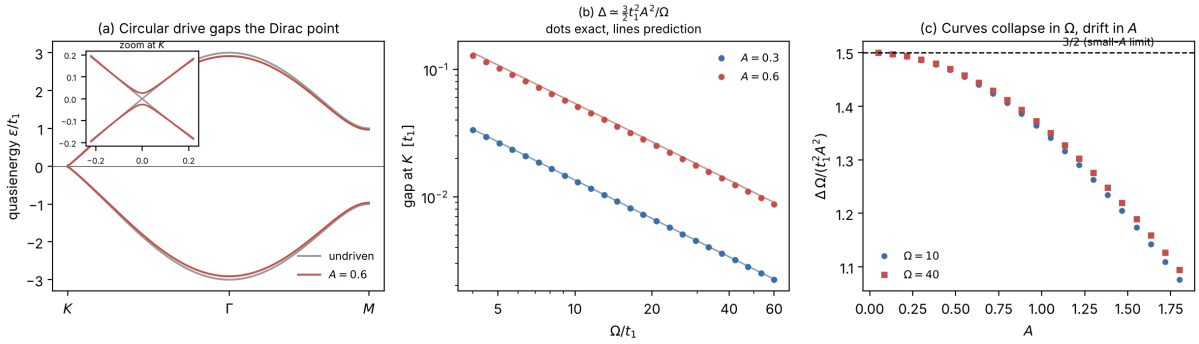


Figure 2: Circularly driven graphene. (a) The drive gaps the Dirac point; the inset zooms on K, where the induced gap is roughly 1% of the bandwidth. (b) The gap follows the high-frequency prediction $\Delta \simeq \frac{3}{2} t_1^2 A^2/\Omega$ over more than a decade in Ω. (c) Two approximations separated: the $\Omega = 10$ and $\Omega = 40$ points collapse onto one another, so the $1/\Omega$ scaling is accurate even at modest frequency, while the coefficient drifts from 3/2 with increasing A — that drift is the $O(A^4)$ correction, not a failure of the Magnus expansion.

The induced gap, measured rather than quoted. Fitting the small-amplitude limit gives

$$\Delta \;\longrightarrow\; \tfrac{3}{2}\, t_1^2 A^2/\Omega \qquad (A \to 0),$$

reproduced to six digits. The coefficient 3/2 is convention dependent — it reflects the choice $|\boldsymbol{\delta}| = 1/\sqrt{3}$ for the neighbour vectors and a dimensionless A — so it is recorded in the code as a measured constant with a test pinning it, rather than copied from a paper using different conventions.

The central check. Matching the induced gap to the static Haldane gap $6\sqrt{3}\, t_2$ yields an effective t_2. The driven system's quasienergy gap and Chern number then agree with the static Haldane model carrying that t_2 to within 2%. The static model of Section 3 is the high-frequency limit of this driven one.

Helicity is not the sign of A. Reversing $A \to -A$ gives the same circulation shifted by half a period, not the opposite handedness, which lives in the relative sign of the two components. The error surfaced as a Chern number that stubbornly refused to flip; helicity is now an explicit parameter, and reversing it flips C while leaving the gap unchanged.

5 Where the Chern number stops working

Rudner, Lindner, Berg and Levin constructed the clean counterexample to the assumption that the preceding picture always holds [4]. Take a checkerboard square lattice and switch on its four

nearest-neighbour bonds one at a time, cyclically around a plaquette, then apply a staggered sublattice potential in a fifth step.

Tune each hopping step to a perfect particle swap, $J\tau = \pi/2$. A bulk particle then hops right, up, left, down — around a closed loop — and lands exactly where it started. The one-period operator is literally the identity,

$$U(T) = \mathbb{1} \quad \text{for every } \mathbf{k} \qquad \Longrightarrow \qquad C = 0 \text{ for both bands,}$$

verified here to 10^{-12}. A particle on the edge cannot complete that loop, because one bond has nothing on the other side; instead of returning home it is shunted one cell along the edge, every period. That is a chiral edge channel.

So the bulk reports "nothing happened" and the edge reports "I moved," and both are correct. The interesting motion occurred mid-period, and the Chern number — built only from $U(T)$ — never looked.

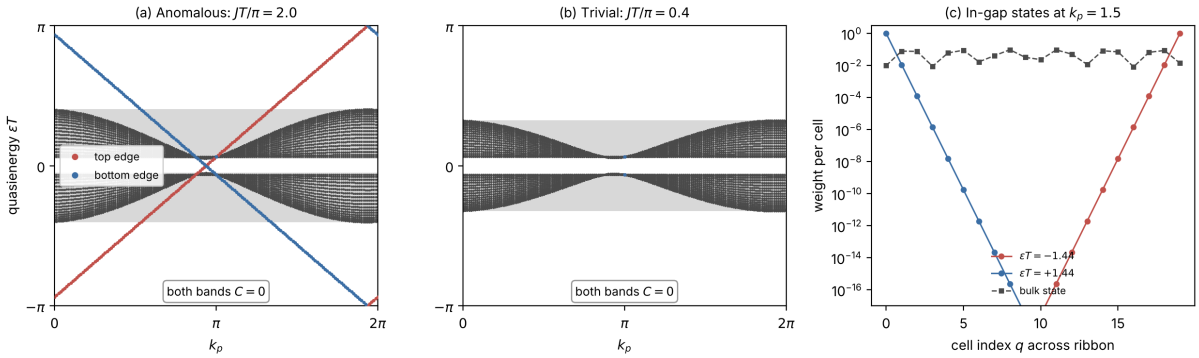


Figure 3: The anomalous Floquet phase. (a) Chiral edge modes, one per edge, sweeping the entire quasienergy zone from $-\pi$ to $+\pi$; grey bands are the projected bulk spectrum. Both bulk bands have $C = 0$. (b) A trivial phase of the same model: also $C = 0$ for both bands, but nothing in the gaps at all. The Chern number takes the same value in both panels and therefore cannot distinguish them. (c) The in-gap states decay exponentially into the bulk over sixteen orders of magnitude while a bulk state stays flat.

Two details that are load-bearing. Both are asserted by tests rather than assumed.

- The cyclic order matters. Right → up → left → down closes the loop; any other ordering leaves a residual $\mathbf{k}$-dependent phase. The test scrambles the order and asserts the loop no longer closes.
- The fifth step is not decoration. With four hopping steps alone the two quasienergy bands touch at $\varepsilon = 0$ for every J, leaving no gap to host an edge mode and no well-defined Chern number. The staggered potential opens it.

Two gaps, not one. A Floquet spectrum lives on a circle, so there is a gap at $\varepsilon = 0$ and one at the zone edge $\varepsilon = \pi$. In Figure 3(a) the edge modes cross both. Checking only the $\varepsilon = 0$ gap would miss half the physics.

Exactness. In each step every site belongs to at most one bond, so the step Hamiltonian is block diagonal in 2×2 blocks whose exponential is closed form. There is no Trotter error and no matrix exponential anywhere in the module; ribbon unitarity holds to 10^{-32}.

6 What is not done

The invariant that does count these modes is a winding number W of the full time-evolution operator $U(\mathbf{k}, t)$ over the $(\mathbf{k}, t)$ torus. It tracks the whole micromotion through the period rather than only the endpoint $U(T)$, which is exactly the information the Chern number discards.

W is not computed here. Figure 3(a) shows its signature — each edge mode winds once around the quasienergy circle as k_p crosses the Brillouin zone, and that winding is what W counts — but the signature is not the invariant. Computing W directly requires a three-dimensional integral over (k_p, k_q, t) using the periodised evolution operator, and is the natural next step.

The further question, and the one that makes the subject more than a curiosity, is what survives disorder: a driven interacting system generically heats to infinite temperature unless localisation intervenes, so Floquet topology and many-body localisation are not independent subjects.

A Validation summary

Every module ships a suite asserting analytically known results. All 37 pass.

Module	Strongest check	Tests
haldane.py	Chern number jumps by 1 exactly at $\lvert M\rvert = 3\sqrt{3}\,t_2\lvert\sin\phi\rvert$	11
floquet.py	Driven quasienergy gap and C match static Haldane with induced t_2	12
anomalous.py	Real-space ring reproduces independent 2×2 Bloch formula to 10^{-16}	14

The last of these is the strongest evidence that the implementation is correct: the ribbon is assembled site-by-site from a bond table in real space, while the bulk comes from an independent analytic 2×2 formula. Closing the ribbon into a ring must reproduce the bulk spectrum exactly, and any error in the bond offsets, the step ordering or the Bloch phases would break the agreement.

References

[1] F. D. M. Haldane, Model for a Quantum Hall Effect without Landau Levels, Phys. Rev. Lett. 61, 2015 (1988).

[2] T. Fukui, Y. Hatsugai, H. Suzuki, Chern Numbers in Discretized Brillouin Zone, J. Phys. Soc. Jpn. 74, 1674 (2005).

[3] T. Oka, H. Aoki, Photovoltaic Hall effect in graphene, Phys. Rev. B 79, 081406(R) (2009).

[4] M. S. Rudner, N. H. Lindner, E. Berg, M. Levin, Anomalous Edge States and the Bulk-Edge Correspondence for Periodically Driven Two-Dimensional Systems, Phys. Rev. X 3, 031005 (2013).

[5] B. A. Bernevig, T. L. Hughes, Topological Insulators and Topological Superconductors, Princeton University Press (2013).